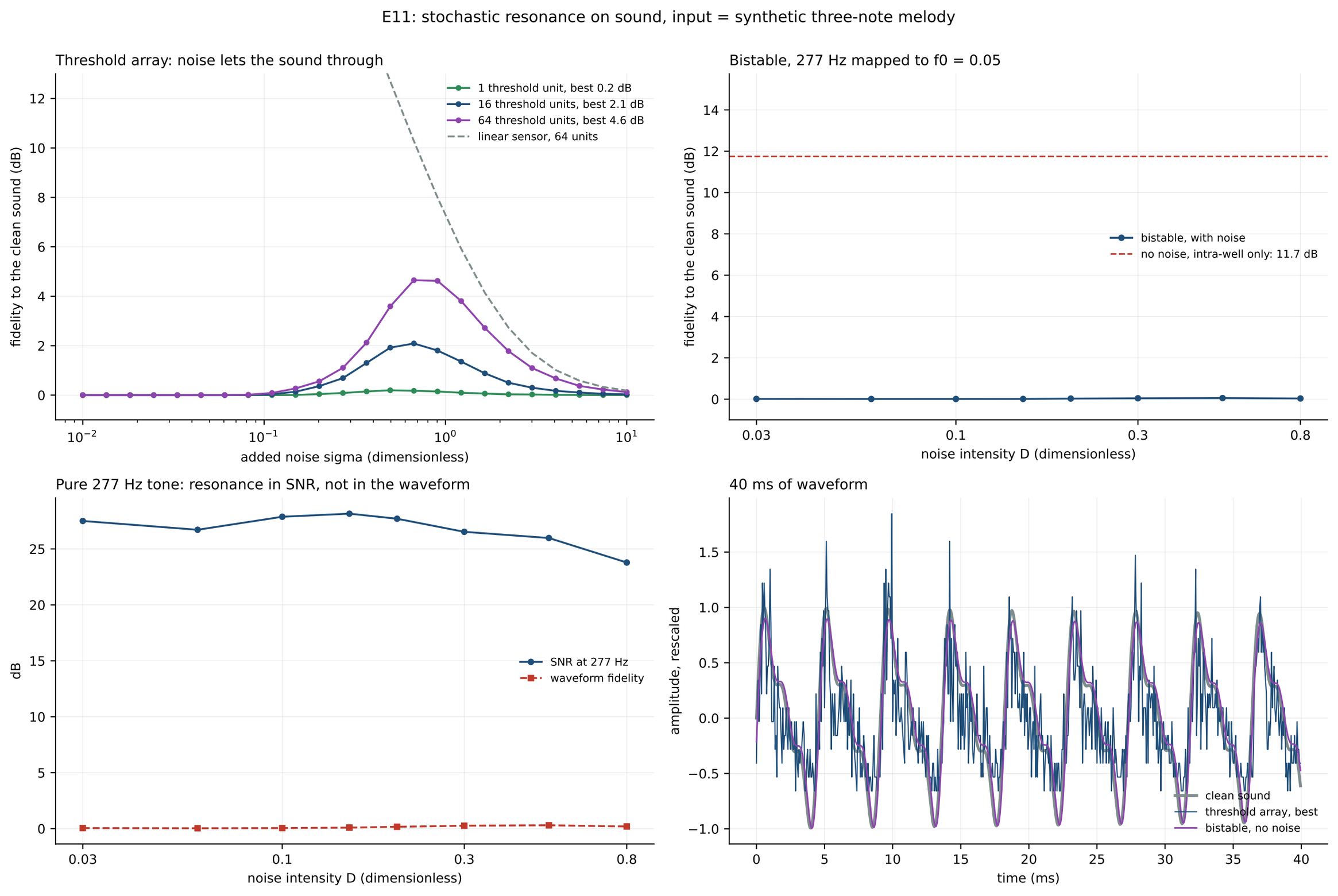
E11: stochastic resonance on sound, input = synthetic three-note melody
Threshold array: noise lets the sound through
1 threshold unit, best 0.2 dB
16 threshold units, best 2.1 dB
64 threshold units, best 4.6 dB
linear sensor, 64 units
fidelity to the clean sound (dB)
added noise sigma (dimensionless)
Bistable, 277 Hz mapped to f0 = 0.05
bistable, with noise
no noise, intra-well only: 11.7 dB
fidelity to the clean sound (dB)
noise intensity D (dimensionless)
Pure 277 Hz tone: resonance in SNR, not in the waveform
SNR at 277 Hz
waveform fidelity
dB
noise intensity D (dimensionless)
40 ms of waveform
clean sound
threshold array, best
bistable, no noise
amplitude, rescaled
time (ms)